

1. rm, users, tree

Aim

To learn file deletion, view logged-in users, and display directory structure. **Algorithm**

1. Open Terminal.
2. Execute the required commands.
3. Observe the output.

Commands

```
rm sample.txt
```

users

tree

Output:

[illegible]

Result

- rm deletes the specified file.
- users displays currently logged-in users.
- tree shows files and folders in a tree format.

2. less, more, vi

Aim

To learn file viewing and editing commands.

Algorithm

1. Open Terminal.
2. Create or use a text file.
3. Execute the commands.

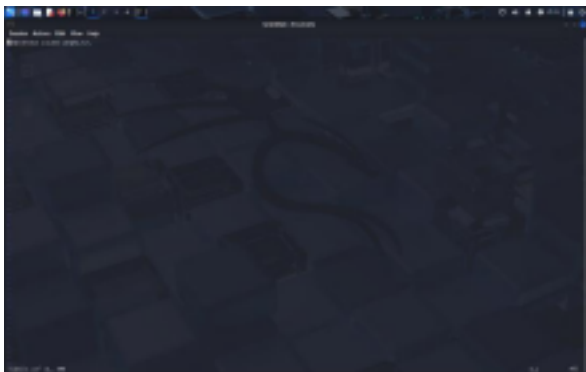
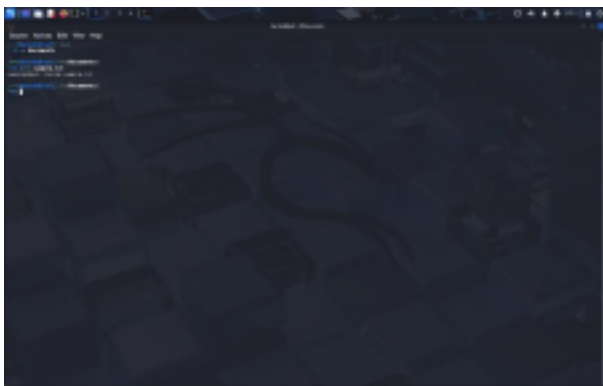
Commands

Less sample.txt

more sample.txt

vi sample.txt

Output



Result

- less views file page by page.
- more displays file one screen at a time.
- vi opens the file in the Vi editor.

3. tree, head, tail

Aim

To display directory structure and view beginning/end of files.

Algorithm

1. Open Terminal.
2. Execute commands.
3. Observe output.

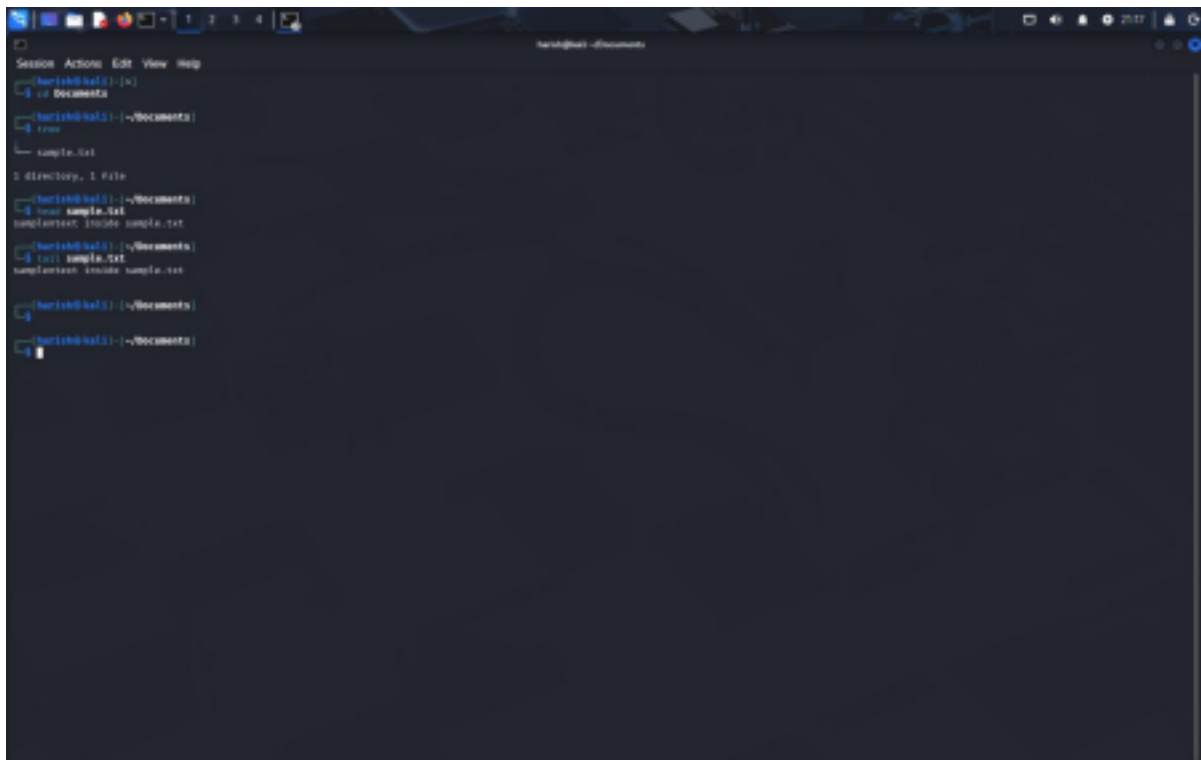
Commands

tree

head sample.txt

tail sample.txt

Output

A screenshot of a terminal window with a dark background. The terminal shows the following commands and their outputs:
1. Command: `tree`
Output: A directory tree showing the current directory (`~/Documents`) containing a subdirectory `tree` and a file `sample.txt`.
2. Command: `head sample.txt`
Output: The first 10 lines of `sample.txt`, which include a header line `0 directory, 1 file` followed by a list of files.
3. Command: `tail sample.txt`
Output: The last 10 lines of `sample.txt`, which are the same list of files as seen in the `head` command output.
The terminal window has a title bar at the top with the text `harish@harish: ~/Documents` and standard window controls.

Result

- tree displays folder hierarchy.
- head shows first 10 lines.
- tail shows last 10 lines.

4. ls, cat, mkdir

Aim

To list files, display file contents, and create directories.

Algorithm

1. Open Terminal.
2. Execute the commands.

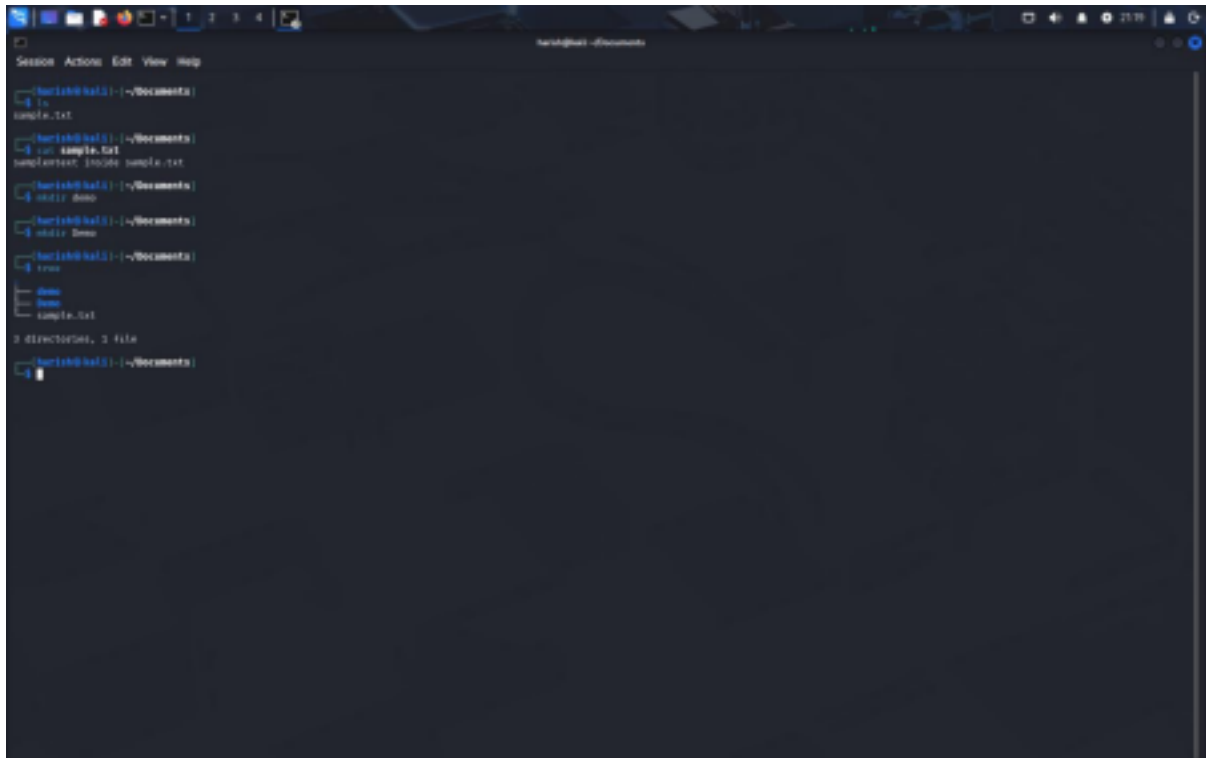
Commands

ls

```
cat sample.txt
```

```
mkdir demo
```

Output:



Result

- `ls` lists files.
- `cat` displays file contents.
- `mkdir` creates a new directory.

5. uptime, date, mv

Aim

To check system uptime, display date, and move/rename files.

Algorithm

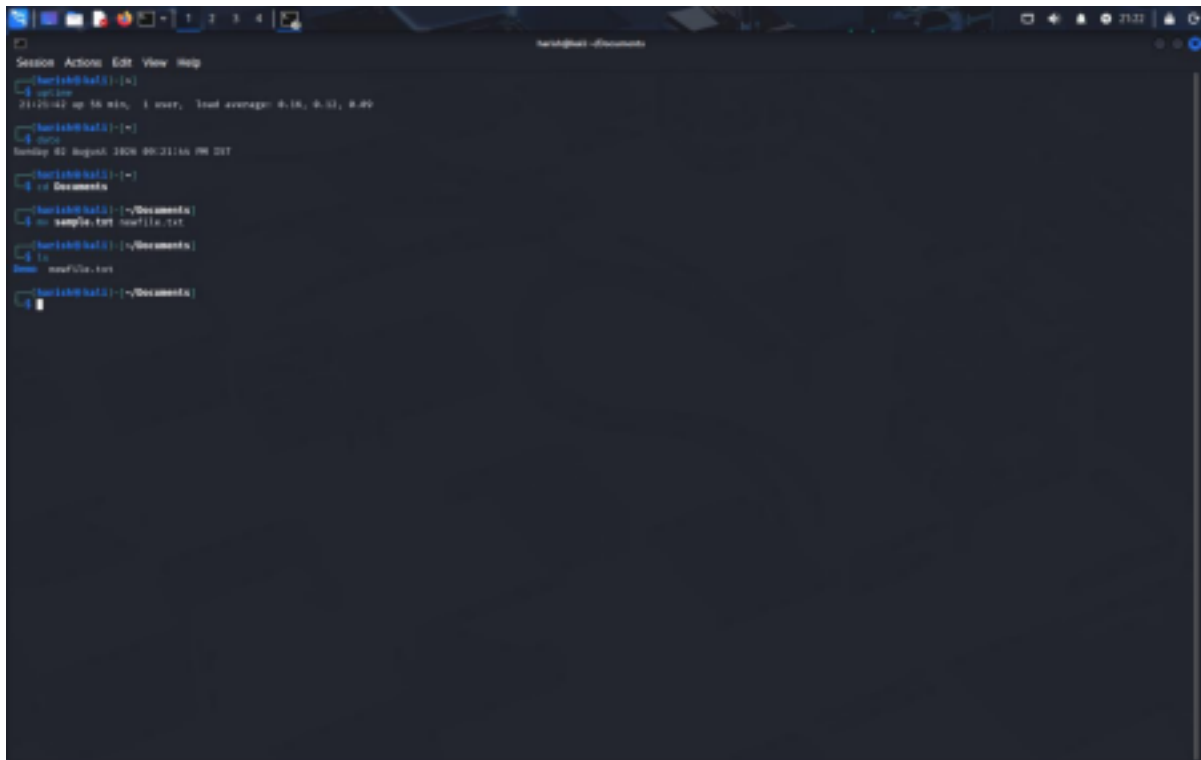
1. Open Terminal.
2. Execute commands.
3. Observe results.

Commands

uptime

date

Output



```
Session Actions Edit View Help
root@kali:~# uptime
 21:20:42 up 56 min,  1 user,  load average: 0.16, 0.32, 0.49
root@kali:~# date
Sunday 30 August 2020 00:21:50 PM IST
root@kali:~# cd Documents
root@kali:~/Documents#
root@kali:~/Documents# touch sample.txt
root@kali:~/Documents# ls
ls
sample.txt
root@kali:~/Documents#
```

Result

- uptime displays system running time.
- date displays current date and time.
- mv moves or renames files.

6. touch, pwd, wget

Aim

To create files, display current directory, and download files.

Algorithm

1. Open Terminal.
2. Execute commands.
3. Verify the output.

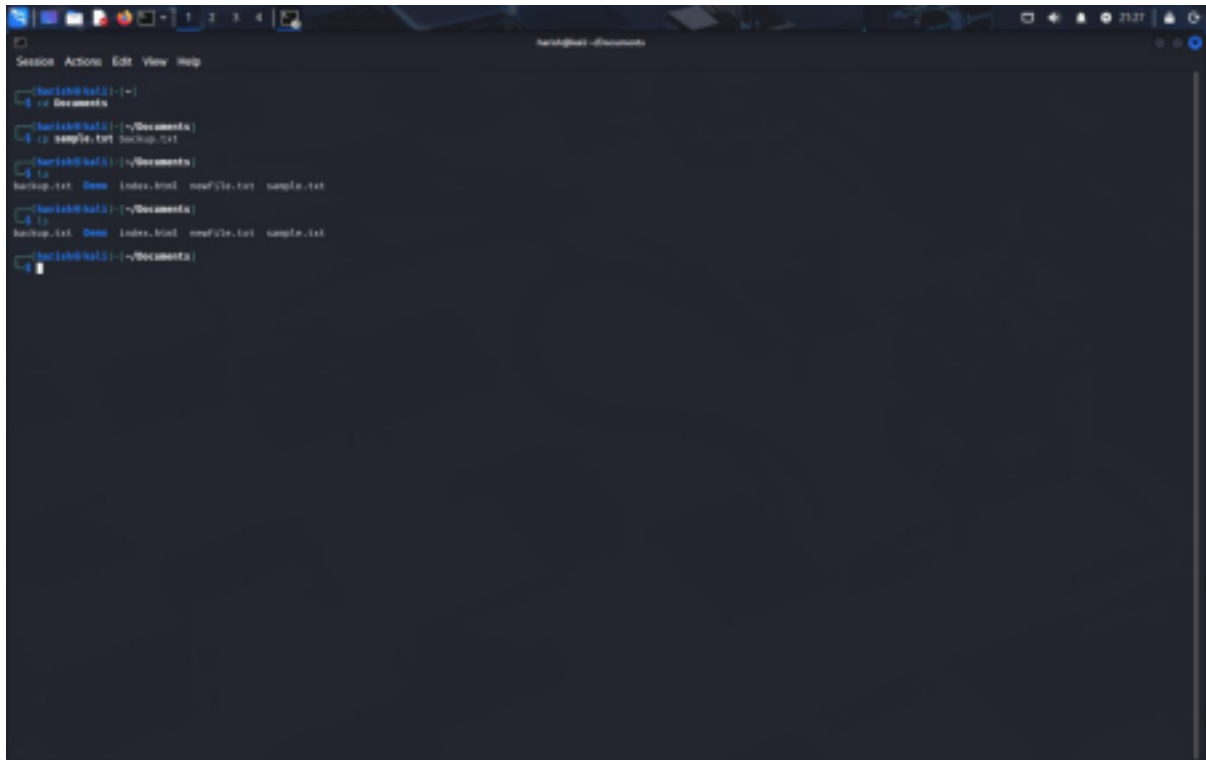
Commands

touch sample.txt

pwd

wget <https://google.com>

Output



```
hari@kali: ~/Documents
hari@kali:~$ cd Documents
hari@kali:~/Documents$ cp sample.txt backup.txt
hari@kali:~/Documents$ ls
backup.txt  Demo  index.html  newFile.txt  sample.txt
hari@kali:~/Documents$ ls
backup.txt  Demo  index.html  newFile.txt  sample.txt
hari@kali:~/Documents$
```

Result

- cd changes the current directory.
- cp copies files.
- ls lists directory contents.

8. Nmap Service Version and OS Detection

Aim

To identify running services, operating system, and perform aggressive scanning. **Algorithm**

1. Open Terminal.
2. Execute Nmap commands.
3. Analyze scan results.

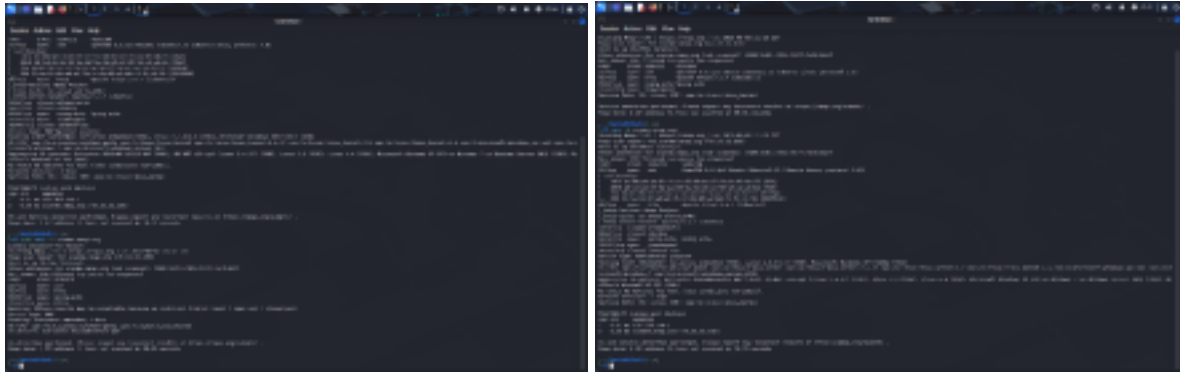
Commands

nmap -sV scanme.nmap.org

nmap -A scanme.nmap.org

sudo nmap -O scanme.nmap.org

Output



Result

- -sV detects service versions.
- -A performs aggressive scan (OS, version, scripts, traceroute).
- -O attempts operating system detection.

9. Nmap Timing and Performance

Aim

To perform scans using different timing templates.

Algorithm

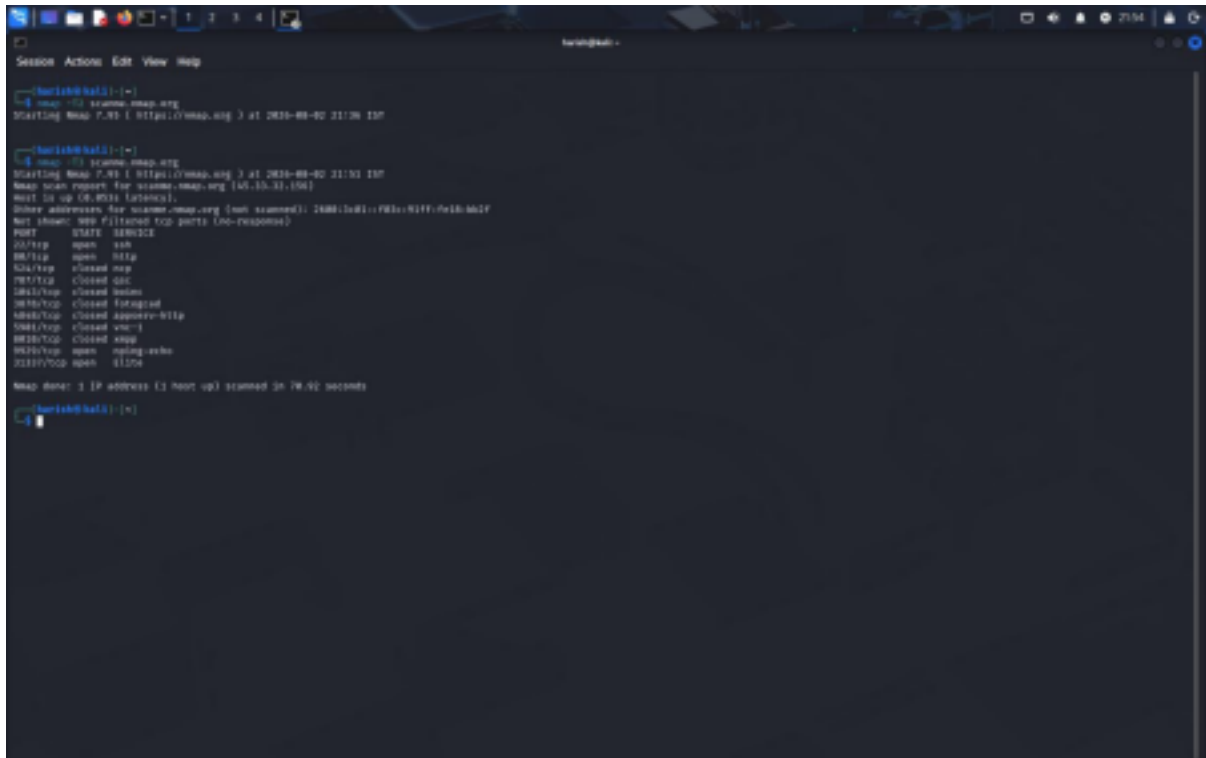
1. Open Terminal.
2. Execute commands.
3. Compare scan speeds.

Commands

```
nmap -T2 scanme.nmap.org
```

```
nmap -T3 scanme.nmap.org
```

Output



Result

- T2 performs a polite (slower) scan, reducing network load.
- T3 performs a normal speed scan.

10. free, sort, history

Aim

To view memory usage, sort data, and display command history.

Algorithm

1. Open Terminal.
2. Execute commands.
3. Observe outputs.

Commands

free -h

```
sort sample.txt
```

history

Output



```
Session Actions Edit View Help
haxim@kali: ~/Documents

haxim@kali:~$ cd Documents
haxim@kali:~/Documents$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.9GiB       1.3GiB       5.9GiB       23Mi       917Mi       6.4GiB
Swap:          2.0GiB         0B       2.0GiB
haxim@kali:~/Documents$ sort sample.txt
quit
name
output
haxim@kali:~/Documents$ history
1  sudo apt update;sudo apt full-upgrade -y
2  sudo apt update;sudo apt install git -y
3  ssh-keygen -t rsa -b 4096 -C "haxim@kali"
4  cd ~/social-engineer-toolkit
5  "[[2024-04-10]]"
6  git clone https://github.com/trustedsec/social-engineer-toolkit.git
7  cd social-engineer-toolkit
8  sudo ./setup.sh
9  cd social-engineer-toolkit
10 cd ~/social-engineer-toolkit
11 cd -
12 git clone https://github.com/trustedsec/social-engineer-toolkit.git
13 sudo su
14 java -version
15 sudo apt install openjdk-13-jdk
16 java -version
17 clear
18 sudo update-alternatives --config java
19 java -version
20 cd Downloads
21 ls
22 sudo apt install ./AMMyn-Setup_amd64.deb
23 clear
24 ammyn
25 ammyn --disable-gpu
26 ammyn
27 ammyn --enable-gpu
28 ammyn
29 sudo apt update ./AMMyn-Setup_amd64.deb
30 sudo apt install ./AMMyn-Setup_amd64.deb
31 cd Downloads
32 sudo apt install ./AMMyn-Setup_amd64.deb
33 ammyn
34 clear
35 ls
36 cd Documents
37 ls
38 mv sample.txt
39 ls
40 -
```

Result

- free -h displays RAM and swap usage.
- sort arranges lines alphabetically.
- history displays previously executed commands.

11. Nmap Aggressive and Insane Timing

Aim

To compare faster Nmap timing templates.

Algorithm

1. Open Terminal.
2. Execute commands.
3. Observe scan duration.

Commands

nmap -T4 scanme.nmap.org

nmap -T5 scanme.nmap.org

Output

A screenshot of a terminal window with a dark background. The terminal shows the execution of the Nmap command 'nmap -T5 scanme.nmap.org'. It displays the Nmap version (7.93), the target IP (192.168.1.108), and the scan progress. The output shows that the host is up and lists several open ports: 22/tcp (ssh), 80/tcp (http), 443/tcp (https), and 4444/tcp (nc). The scan is completed in 82.75 seconds. The terminal window has a title bar with 'Terminal' and standard window controls.

```
Session Actions Edit View Help
[harish@kali:~]
[~] nmap -T5 scanme.nmap.org
Starting Nmap 7.93 ( https://nmap.org ) at 2020-08-02 21:58 IST
Nmap scan report for scanme.nmap.org [192.168.1.108]
Host is up (0.0000 latency).
Other addresses for scanme.nmap.org (not scanned): 192.168.1.108::FW::Niff-Feld-Mirf
Not shown: 995 filtered TCP ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
4444/tcp   open  nc
Nmap done: 1 IP address (1 host up) scanned in 82.75 seconds

[harish@kali:~]
[~] nmap -T5 scanme.nmap.org
Starting Nmap 7.93 ( https://nmap.org ) at 2020-08-02 21:58 IST
Nmap scan report for scanme.nmap.org [192.168.1.108]
Host is up (0.0000 latency).
Other addresses for scanme.nmap.org (not scanned): 192.168.1.108::FW::Niff-Feld-Mirf
Not shown: 995 filtered TCP ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
4444/tcp   open  nc
Nmap done: 1 IP address (1 host up) scanned in 82.75 seconds

[harish@kali:~]
```

Result

- -T4 performs an aggressive speed scan.
- -T5 performs an insane speed scan (may miss results on slower networks).

12. Port Scanning Tools

Aim

To identify open ports on a target system.

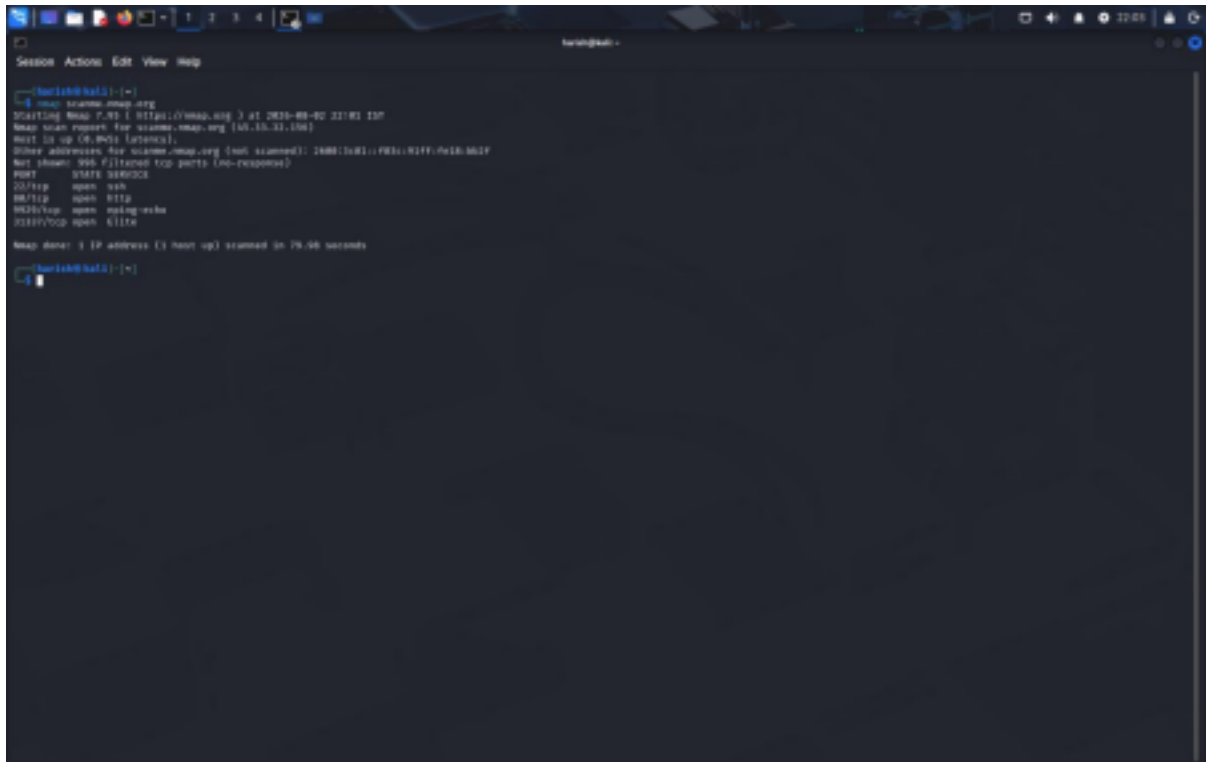
Algorithm

1. Open Terminal.
2. Execute port scanning command.
3. Analyze open ports.

Command

nmap scanme.nmap.org

Output



```
burish@kali:~$ nmap -sn 192.168.1.0/24
Nmap scan report for 192.168.1.0/24
Host is up (0.0000s latency).
Other addresses for 192.168.1.0/24 (not scanned): 192.168.1.1-192.168.1.254
Host down: 999.0000s (no response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
5432/tcp   open  postgresql
```

Result

The command lists open TCP ports and their status.

13. Host Discovery

Aim

To identify active hosts in a network.

Algorithm

1. Open Terminal.
2. Execute host discovery command.
3. Observe active hosts.

Command

```
nmap -sn 192.168.1.0/24
```

Output



Result

Displays all active hosts in the specified subnet.

14. Cracking Password Using Hydra (Lab Environment Only)

Aim

To demonstrate password auditing using Hydra against an authorized test

target. **Algorithm**

1. Open Terminal.
2. Provide username and password list.
3. Execute Hydra.

Command

```
hydra -l admin -P passwords.txt ssh://192.168.1.10
```

Output

Hydra is a password auditing tool designed to test authentication services such as SSH, FTP, HTTP, and Telnet. Executing Hydra requires an **authorized target system** with a running login service and a valid username/password wordlist. Since no dedicated lab target was available, **this experiment was not executed to avoid performing unauthorized password attacks against external systems.** The experiment should be performed only in a controlled lab environment.

**Result**

Hydra attempts authentication using the supplied wordlist and reports valid credentials if found on the authorized test system.

15. Information Gathering Using theHarvester**Aim**

To collect publicly available information about a domain.

Algorithm

1. Open Terminal.
2. Execute theHarvester.
3. Review collected information.

Command

```
theHarvester -d example.com -b bing
```

Output

**Result**

TheHarvester collects publicly available information such as email addresses, subdomains, hosts, and other OSINT data related to the target domain.

16. Use Google & Who.is for Reconnaissance

Aim: To perform reconnaissance using Google search and Whois.

Algorithm:

Open a web browser.

Perform Google search queries.

Execute the Whois lookup.

Observe the collected information.

Output:



Result:

Reconnaissance was successfully performed using Google and Whois.

17. Windows OS Commands Execution (tracert, ping, ipconfig, netstat)

Aim:

To execute basic Windows networking commands.

Algorithm:

Open Command Prompt.

Execute tracert command.

Execute ping command.

Execute ipconfig command.

Execute netstat command.

Observe the outputs.

Output:



Result:

Windows networking commands were executed successfully.

18. Vulnerabilities Analysis Using CGI Scanning with Nikto

Aim:

To perform vulnerability analysis using the Nikto scanner.

Algorithm:

Open the Kali Linux terminal.

Execute the Nikto scan command.

Observe the detected vulnerabilities.

Record the findings.

Output:



Result:

The vulnerability analysis was successfully performed using Nikto.

19. Wireshark Sniffer for Network Traffic & Analysis

Aim:

To capture and analyze network packets using Wireshark.

Algorithm:

Open Wireshark.

Select the active network interface.

Start packet capture.

Generate network traffic.

Apply protocol filters.

Analyze the captured packets.

Stop packet capture.

Output:

**Result:**

Network traffic was successfully captured and analyzed using Wireshark.

20. Implement the Boot Sector Virus (msfvenom)**Aim:**

To generate a payload using msfvenom.

Algorithm:

Open the Kali Linux terminal.

Execute the msfvenom command.

Generate the payload.

Observe the generated file.

Output:

**Result:**

The payload was successfully generated using msfvenom.

21. Batch File Execution**Aim:**

To create and execute a batch file in Windows.

Algorithm:

1. Open Notepad.
2. Write the batch commands.
3. Save the file with a .bat extension.
4. Execute the batch file.
5. Observe the output.

Output:

**Result:**

The batch file was successfully executed.

22. Packet Analyser Tool**Aim:**

To analyze network packets using a packet analyzer tool.

Algorithm:

Open Wireshark.

Capture network packets.

Apply protocol filters.

Analyze packet details.

Stop the capture.

Output:

**Result:**

Network packets were successfully analyzed.

23. Port Scanning Tools (Extended)**Aim:**

To perform extended port scanning using Nmap.

Algorithm:

Open the Kali Linux terminal.

Execute the extended port scan command.

Observe the scan results.

Record the open ports.

Output:

**Result:**

Extended port scanning was successfully performed.

24. Nmap Timing & Performance**Aim:**

To perform timing and performance scans using Nmap.

Algorithm:

Open the Kali Linux terminal.

Execute timing scan commands.

Observe the scan results.

Record the output.

Output:

**Result:**

Nmap timing and performance scans were successfully executed.

25. Using Shell Programming

- i) Read a number and find its square
- ii) Find the biggest of three numbers

(i): Shell Programming – Find Square of a Number

Aim:

To write a shell program to calculate the square of a number.

Algorithm:

Open the Kali Linux terminal.

Create a shell script.

Read a number from the user.

Calculate its square.

Display the result.

Output:



Result:

The shell program successfully calculated the square of the given number.